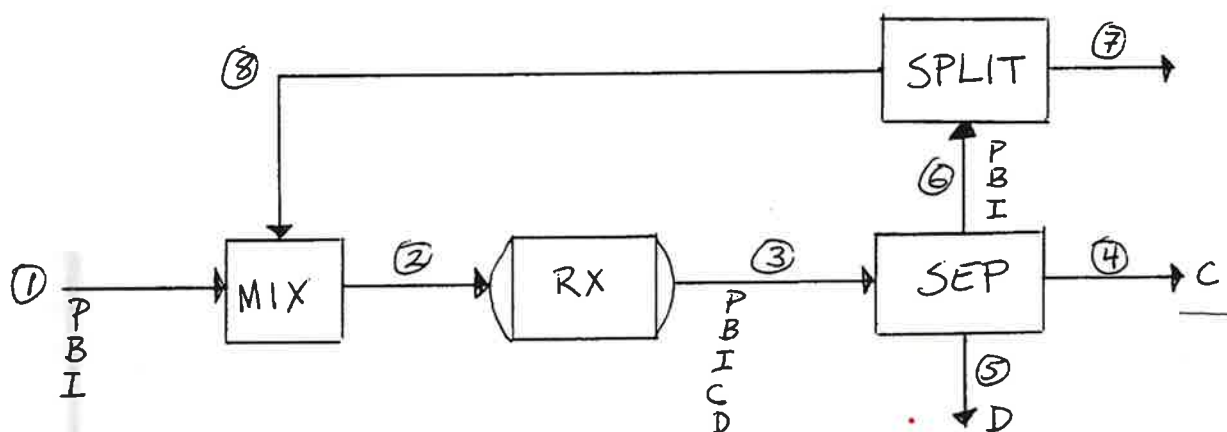


To receive full credit, you must show all of your work.

- (1) (60 pts.) Cumene (C_9H_{12}) is synthesized from propylene (C_3H_6) and benzene (C_6H_6):



The undesired byproduct diisopropylbenzene ($C_{12}H_{18}$) is produced in a second competing reaction. Refer to the block flow diagram below. The fresh feed contains 95 mol/s propylene, 90 mol/s benzene, and 5 mol/s of propane I (C_3H_8) which is inert. This stream is combined in the mixer with the recycle stream. The reactor single-pass conversion is 80% of the propylene and 72% of the benzene. The reactor product goes to a separator which has three output streams: a stream of pure cumene, a stream of pure diisopropylbenzene, and a stream containing the unreacted propylene, benzene, and propane, which is sent to a splitter. The splitter purges 10% of this stream and recycles the remainder.



- (a) Finish labeling the flow diagram above. In the column "Entire Process" (see Table below), use the degree of freedom analysis (see p. 105 of Murphy) to determine if the problem is well posed. Give a brief explanation for each entry. (10 pts.)

	Entire Process	Explanation
4a # stream variables	40	There are 8 streams and 5 components (P, B, I, C, D). Each component flowrate in each stream is a variable $\rightarrow 8 \times 5 = 40$
4b # reaction variables	2	There are 2 independent reactions, so there are 2 extent of reaction variables
# indep variables	42	Total independent variable = stream variable + reaction variable $\rightarrow 40 + 2 = 42$
5a # specified flows	3	The fresh feed stream specifies flowrates of P, B, I
5b # spec stream comps and split	1 $+ (3-1)(2-1)$	The splitter specifies that 10% of the stream is purged, which gives one independent split fraction relation
5c # performance specs	2	The reactor conversion is specified: 80% conversion of propylene and 72% conversion of benzene
5d # material balances	5	There are 5 components in the system $\rightarrow$ 5 independent material balance equations
# indep eqns	11	Total independent equation = 3 (specified flow) + 1 (split) + 2 (performance) + 5 (material balance) = 11
DoF	31	DoF = variables - equations = 42 - 11 = 31 $\rightarrow$ DoF $\neq 0 \rightarrow$ the process is not well posed

Problem 2 continued

- (b) Calculate the fractional yield for cumene for the following percent excesses in benzene: 0%, 100%, and 400%. Be careful to avoid roundoff error. (15 pts.)

$$\text{use: } K = \frac{4Y^2}{(1-Y)(1+E-Y)} \quad K = 0.0005$$

Case 1: 0% excess benzene

$$\begin{aligned} \Rightarrow E &= 0 \\ \Rightarrow 0.0005 &= \frac{4Y^2}{(1-Y)(1-Y)} \\ \Rightarrow 0.0005(1-Y)^2 &= 4Y^2 \\ \Rightarrow Y &= 0.01106 \end{aligned}$$

Case 2: 100% excess benzene

$$\begin{aligned} \Rightarrow E &= 1 \\ \Rightarrow 0.0005 &= \frac{4Y^2}{(1-Y)(2-Y)} \\ \Rightarrow Y &= 0.01563 \end{aligned}$$

Case 3: 400% excess benzene

$$\begin{aligned} \Rightarrow E &= 4 \\ \Rightarrow 0.0005 &= \frac{4Y^2}{(1-Y)(5-Y)} \Rightarrow Y = 0.02463 \end{aligned}$$

% Excess Benzene	0%	100%	400%
Fractional Yield	0.01106	0.01563	0.02463

- (c) Suggest how the design of the process could be changed to improve the fractional yield of cumene. (5 pts.)

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Problem 1 continued

(b) Write the appropriate balance equations. Calculate the molar flow rates in the process in mol/s. Be sure to define the system for any sets of balance equations you write! Fill in the Table below with your results. (45 pts.)

Let reactor extents be ξ_1 for R₁ and ξ_2 for R₂ | $P_2 = P_1 + P_8 = 0.95 + 0.18P_2$

From stoichiometry:

$$I_3 = P_2 - \xi_1 - \xi_2$$

$$D_3 = D_2 - \xi_1$$

$$I_3 = I_2$$

$$C_3 = \xi_1 - \xi_2$$

$$D_3 = \xi_2$$

Given single-pass conversion:

$$\xi_1 + \xi_2 = 0.8$$

$$P_2$$

$$\frac{\xi_1}{D_2} = 0.72$$

$$D_2$$

$$\Rightarrow \xi_1 = 0.72 D_2$$

$$\Rightarrow \xi_1 + \xi_2 = 0.80 P_2$$

$$\Rightarrow P_3 = 0.20 P_2$$

$$D_3 = 0.28 D_2$$

$$I_3 = I_2$$

splitter relations

10% purge, 90% recycle:

$$P_7 = 0.10 P_6 \quad P_8 = 0.9 P_6$$

$$D_7 = 0.10 D_6 \quad D_8 = 0.9 D_6$$

$$I_7 = 0.10 I_6 \quad I_8 = 0.9 I_6$$

Single separator sends all unreacted

P, B, I to stream 6:

$$P_3 = P_6, D_6 = D_3, I_6 = I_3$$

$$\Rightarrow P_8 = 0.9 P_3 = 0.9 (0.20 P_2) = 0.18 P_2$$

$$\Rightarrow D_8 = 0.9 D_3 = 0.9 (0.28 D_2) = 0.252 D_2$$

$$I_8 = 0.9 I_3 = 0.9 I_2$$

$$0.82 P_2 = 95$$

$$\Rightarrow P_2 = 115.8537$$

$$P_2 = D_1 + D_8 = 90 + 0.252 D_2$$

$$\Rightarrow 0.748 D_2 = 90 \Rightarrow D_2 = 120.32$$

$$I_2 = I_1 + I_8 = 5 + 0.9 I_2$$

$$\Rightarrow 0.10 I_2 = 5 \Rightarrow I_2 = 50$$

$$\xi_1 = 0.72 D_2 = 0.72 (120.32) = 86.6310$$

$$\xi_1 + \xi_2 = 0.80 P_2 = 0.80 (115.8537) = 92.6829$$

$$\Rightarrow \xi_2 = 92.6829 - 86.6310 = 6.0519$$

$$P_3 = 0.20 P_2 = 23.1707$$

$$D_3 = 0.28 D_2 = 33.6898$$

$$I_3 = 50$$

$$C_3 = \xi_1 - \xi_2$$

$$= 86.6310 - 6.0519$$

$$= 80.5791$$

Stream Summary Table: All flows in mol/s

↓/Stream→	1	2	3	4	5	6	7	8
P	95	115.8537	23.1707	-----	-----	23.1707	2.3171	20.8537
B	90	120.32	33.6898	-----	-----	33.6898	3.3690	30.3209
I	5	50	50	-----	-----	50	5	45
C	-----	-----	80.5791	80.5791	-----	-----	-----	-----
D	-----	-----	6.0519	-----	6.0519	-----	-----	-----

(c) Calculate the overall fractional conversions of propylene, the fractional selectivity, and the fractional yield (5 pts.)

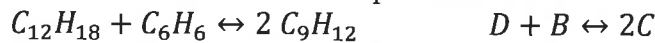
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NAME Tran Ngoc Trinh

(2) (40 pts.) In the production of cumene, diisopropylbenzene is produced as a byproduct. To avoid wasting this material it can be converted to cumene in an equilibrium reaction:



A stream of 10 mol/s diisopropylbenzene is mixed with excess benzene and fed to an equilibrium reactor. The relation between the equilibrium constant and the mole fractions in the product stream is:

$$K = \frac{x_C^2}{x_D x_B}$$

where under the reactor conditions $K = 0.0005$, which is a small value favoring the reactants.

(a) Write the appropriate species balance equations for this reaction. Combine these results with the equilibrium constant relation to derive a general expression for the percent excess benzene. (20 pts.)

a) Let ξ = extent of reaction

$$D = 10 - \xi$$

$$B = B_0 - \xi$$

$$C = 2\xi$$

total outlet moles:

$$N = (10 - \xi) + (B_0 - \xi) + 2\xi$$

$$\Rightarrow N = 10 + B_0$$

$$x_D = \frac{10 - \xi}{10 + B_0}$$

$$x_B = \frac{B_0 - \xi}{10 + B_0}$$

$$x_C = \frac{2\xi}{10 + B_0}$$

$$K = \frac{x_C^2}{x_D x_B}$$

$$\Rightarrow K = \frac{\left(\frac{2\xi}{10 + B_0}\right)^2}{\left(\frac{10 - \xi}{10 + B_0}\right)\left(\frac{B_0 - \xi}{10 + B_0}\right)}$$

stoichiometric benzene needed for 10 mol/s D is 10 mol/s

Let fractional excess benzene be E :

$$B_0 = 10(1 + E)$$

$$\text{where: } E = \frac{\% \text{ excess benzene}}{100}$$

$$\Rightarrow K = \frac{4\xi^2}{(10 - \xi)(10(1 + E) - \xi)}$$

For cumene, maximum possible product is 20 mol/s, so:

$$Y = \frac{2\xi}{20} = \frac{\xi}{10} \Rightarrow \xi = 10Y$$

$$\Rightarrow K = \frac{4(10Y)^2}{(10 - 10Y)(10(1 + E) - 10Y)}$$

$$\Rightarrow K = \frac{4Y^2}{(1 - Y)(1 + E - Y)}$$

$$\Rightarrow 1 + E - Y = \frac{4Y^2}{K(1 - Y)}$$

$$\Rightarrow E = Y - 1 + \frac{4Y^2}{K(1 - Y)}$$

$$\Rightarrow \% \text{ excess benzene} = 100 \left[Y - 1 + \frac{4Y^2}{K(1 - Y)} \right]$$

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Problem # 1

NAME: Tran Ngoc Trinh

Continue:

$$D_3 = E_2 = 6.0519$$

Pure cumene:

$$C_4 = C_3 = 80.5791$$

Pure diisopropylbenzene:

$$D_5 = P_3 = 6.0519$$

unreacted stream:

$$P_6 = P_3 = 23.1707$$

$$B_6 = B_3 = 33.6898$$

$$I_6 = I_3 = 50$$

Large stream 7:

$$P_7 = 0.10(23.1707) = 2.3171$$

$$B_7 = 0.10(33.6898) = 3.369$$

$$I_7 = 0.10(50) = 5$$

Recycle stream 8:

$$P_8 = 0.9(23.1707) = 20.8537$$

$$B_8 = 0.9(33.6898) = 30.3209$$

$$I_8 = 0.9(50) = 45$$

$$c) X_P = \frac{P_{in} - P_{out}}{P_{in}}$$

$$\rightarrow X_P = \frac{95 - 2.3171}{95} = 0.9756$$

$$S = \frac{\text{desired product}}{\text{undesired product}} = \frac{C_4}{D_5} = \frac{80.5791}{6.0519} = 13.3158$$

fractional selectivity

Since 1 mol P makes 1 mol C, and 1 mol D uses 2 mol P:

$$Y = \frac{C_4}{C_4 + 2D_5} = \frac{80.5791}{80.5791 + 2(6.0519)}$$

$$\Rightarrow Y = 0.8694 \quad \text{why 2?}$$

